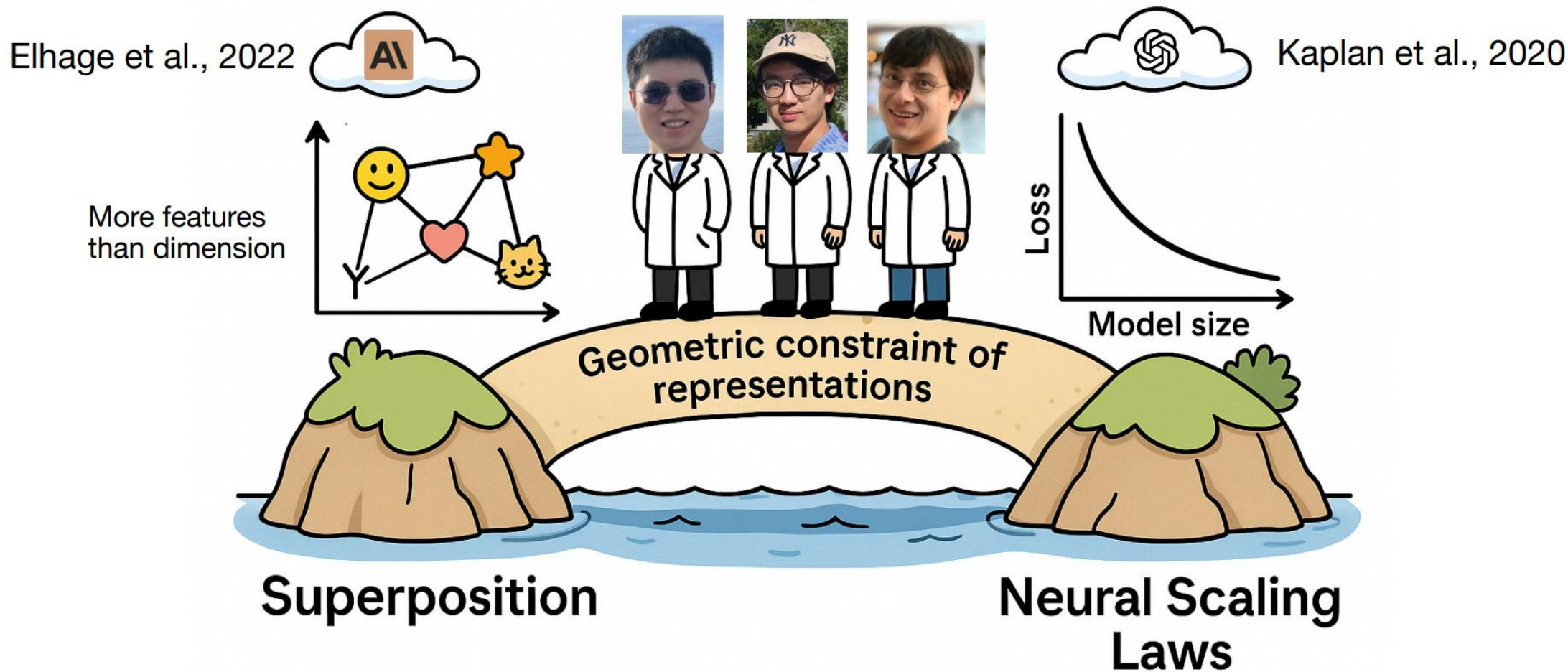
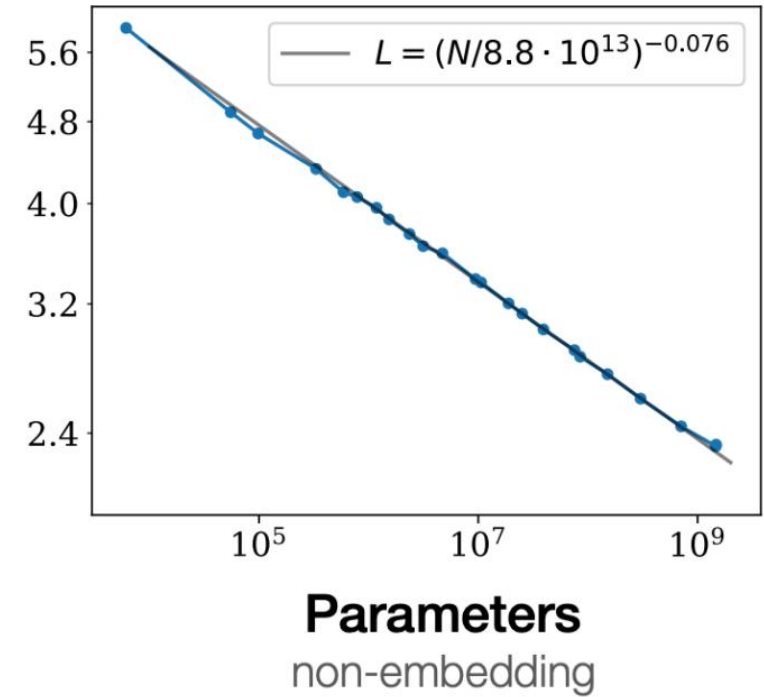
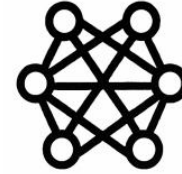
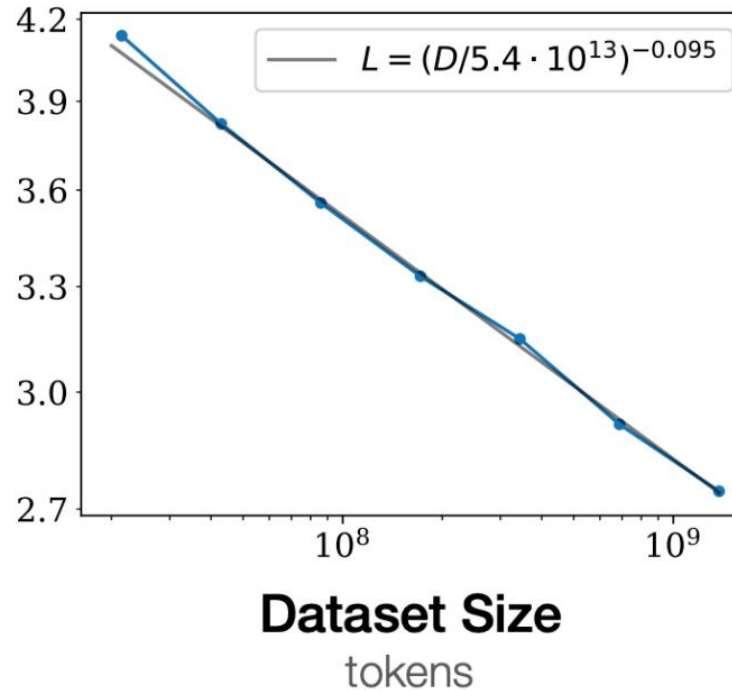
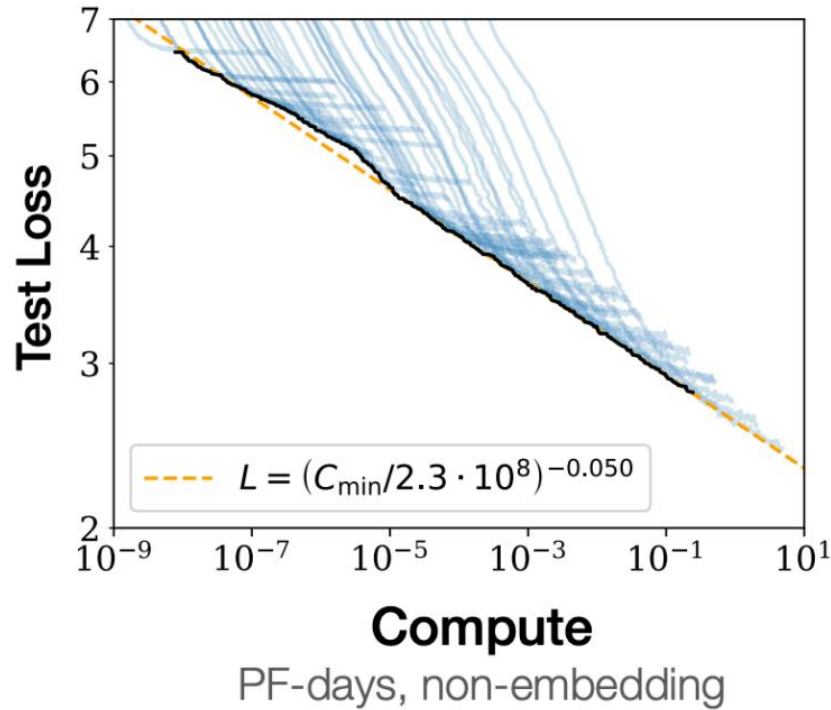


Superposition Yields Robust Neural Scaling

Yizhou Liu Ziming Liu Jeff Gore



Why are large language models **large**?

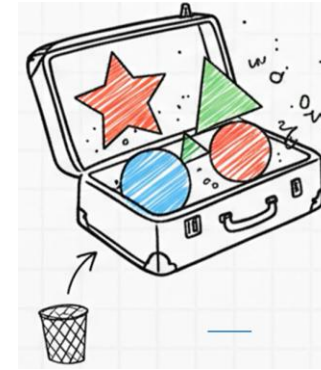


Kaplan et al., 2020

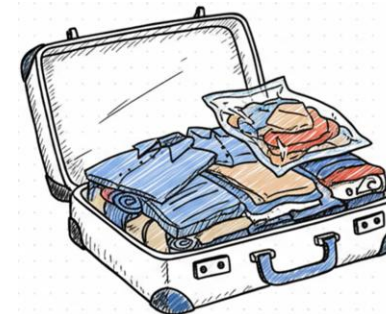
The Core Problem

- Too Many Features, Too Little Space

- **Scenario A (No Superposition):** You carefully fold just m items and put them in. You leave the rest behind.



- **Scenario B (Superposition):** You jam *everything* in. The clothes get wrinkled and overlap, but you carry everything.

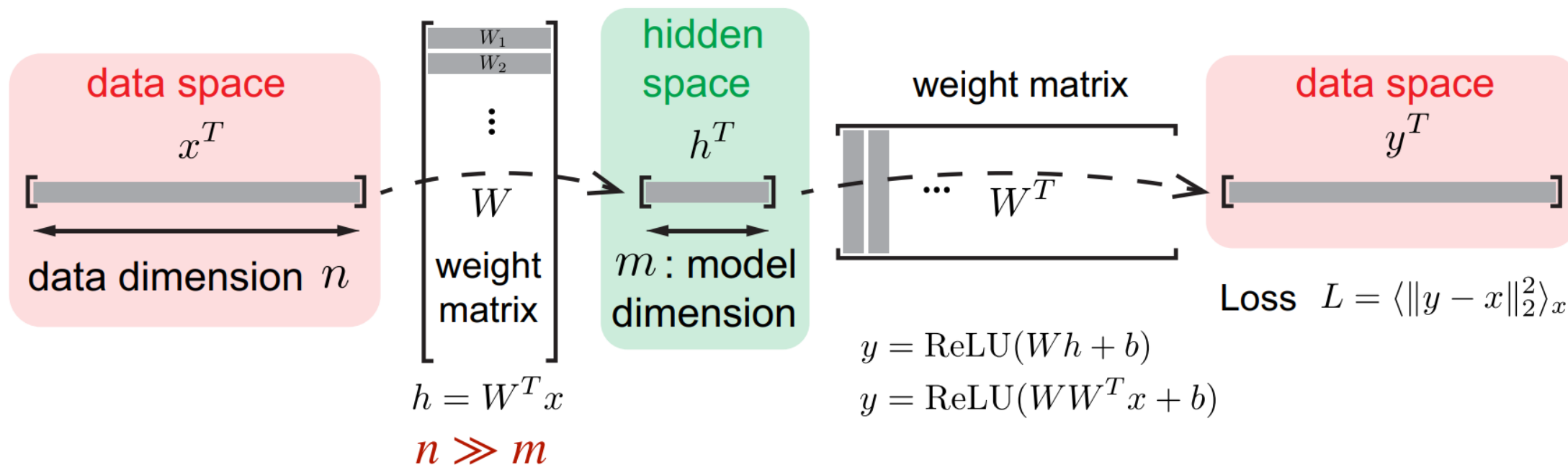


LLM

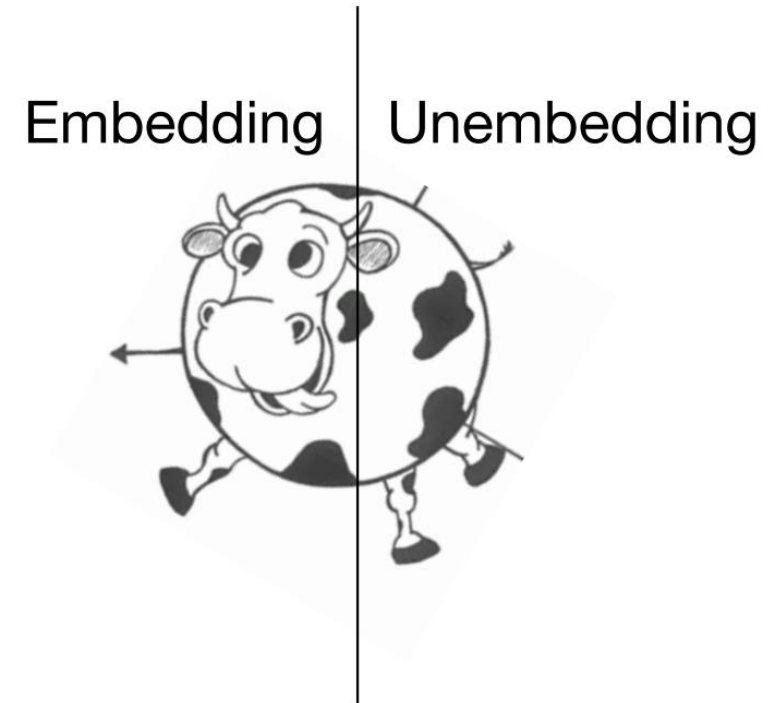
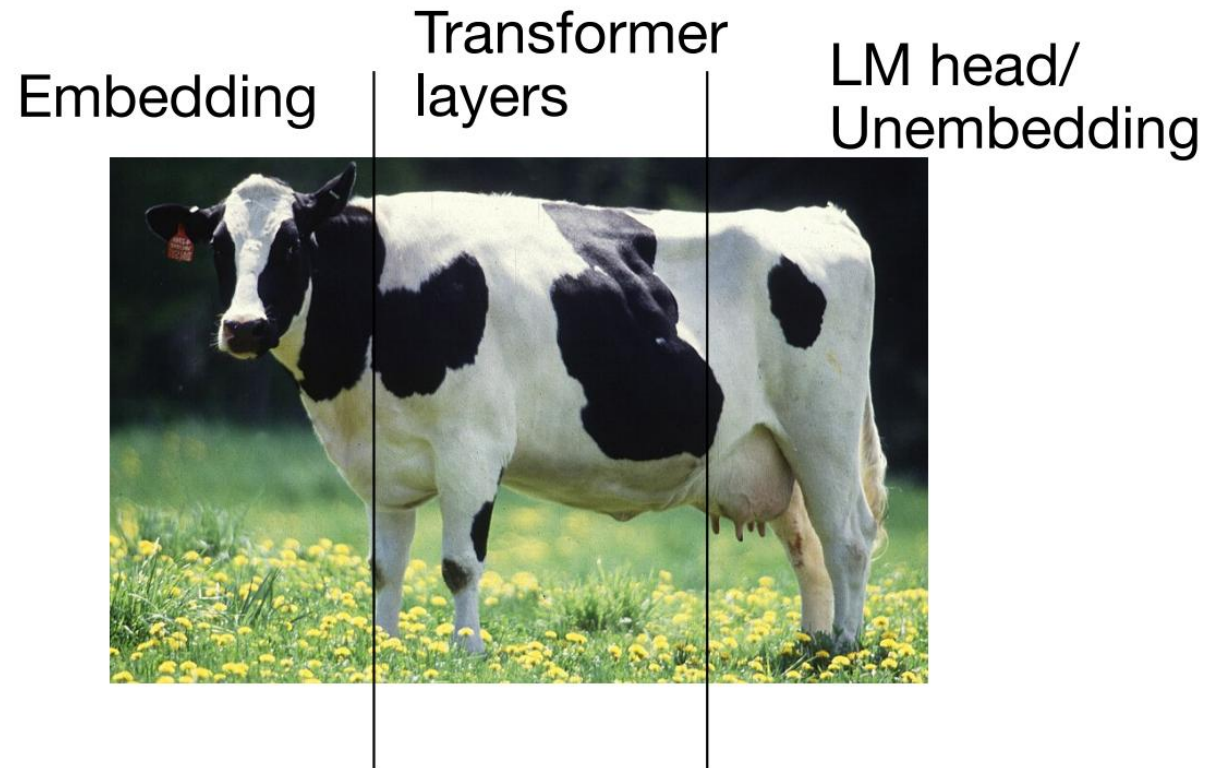
- The vocabulary of English is massive ($n \approx 50,000+$).
- The hidden dimension of a model is much smaller ($m \approx 4,000$ to $12,000$)

Toy model

Elhage et al., 2022

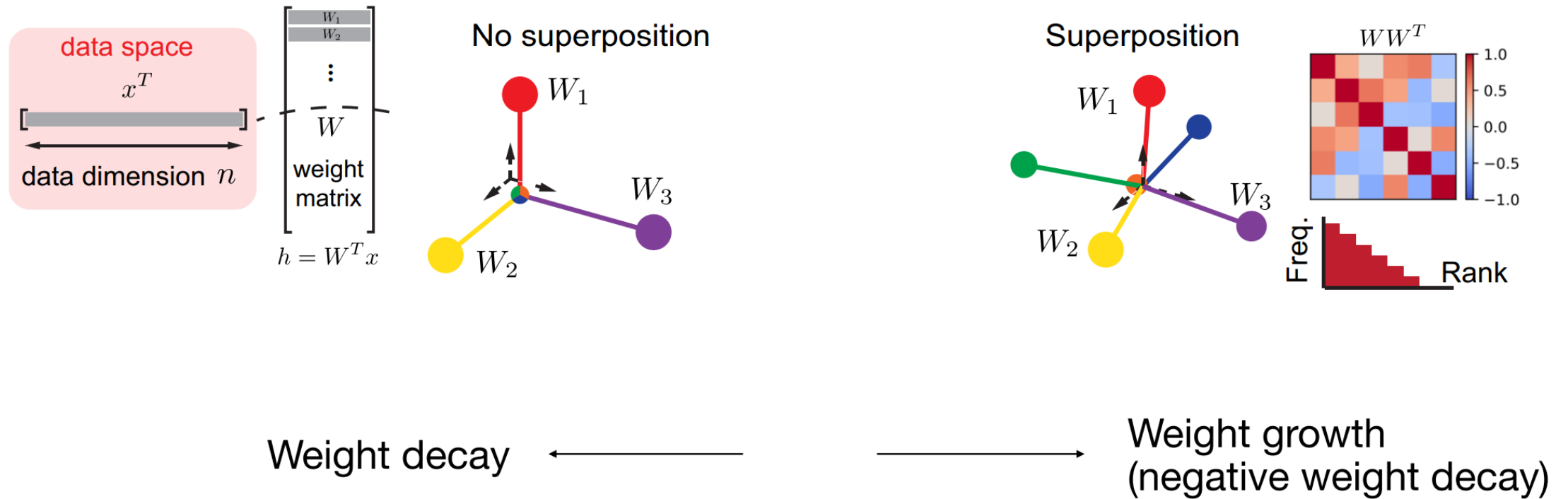


LLM vs Toy Model



Control Superposition

Weight decay



Weak Superposition

$$L \approx \sum_{i=m+1}^n p_i$$

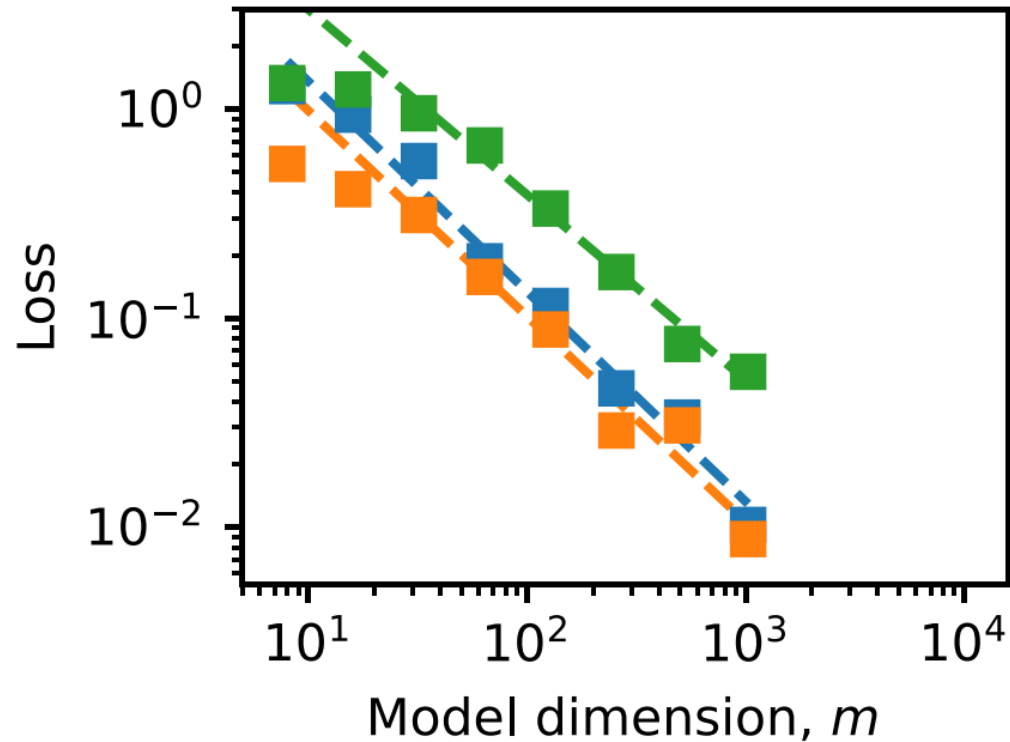
The error comes purely from the **unlearned features**.

The takeaway:

In weak superposition, the scaling depends entirely on the data.

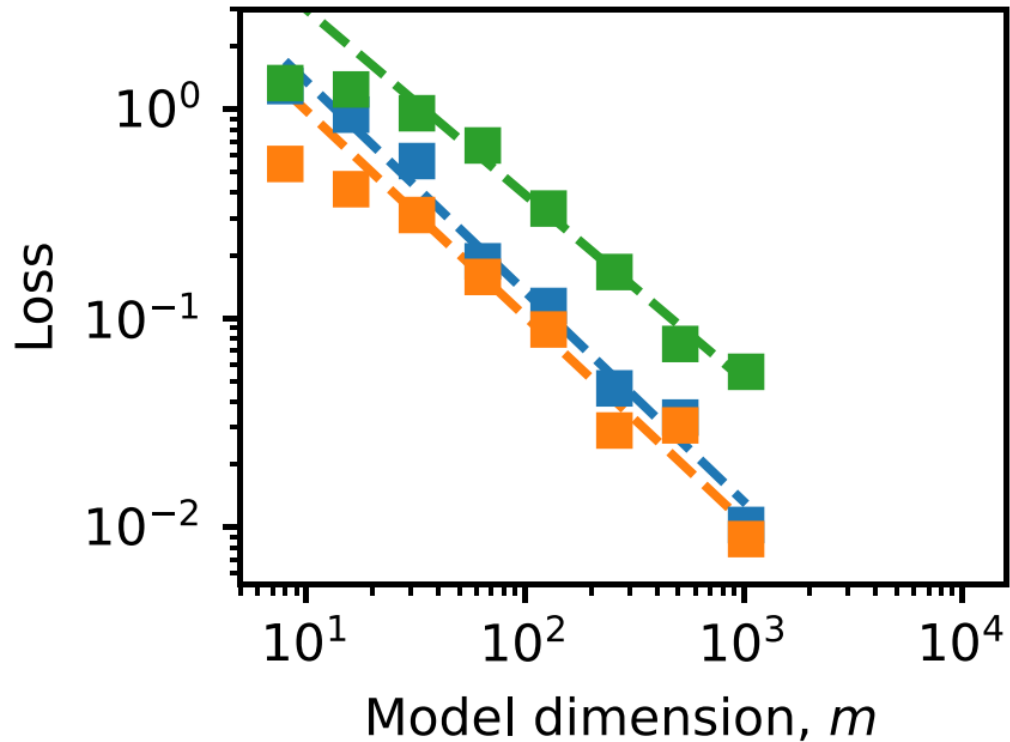
Strong Superposition

Loss scales inversely with the model width ($1/m$), **regardless of the data distribution.**



Strong Superposition

Loss scales inversely with the model width ($1/m$), **regardless of the data distribution.**



Why?

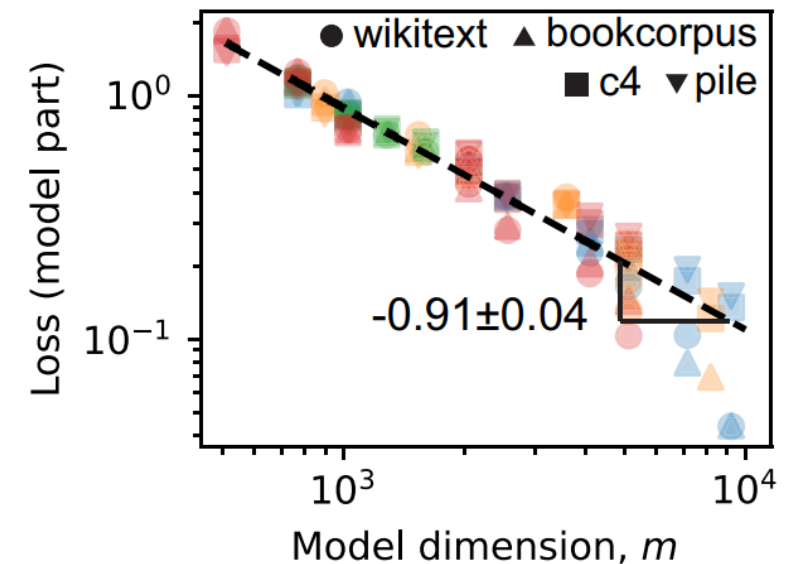
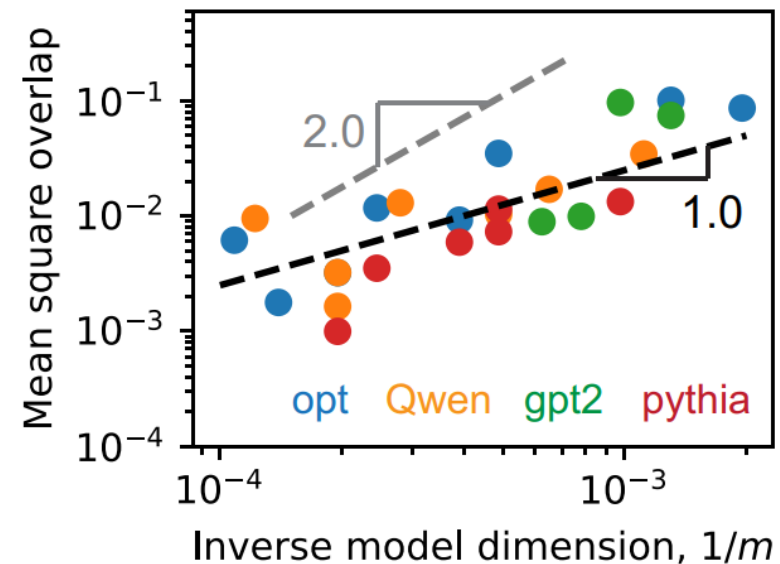
$$\text{Mean Squared Overlap} \propto \frac{1}{m}$$

$$L \propto \frac{1}{m}$$

Are LLMs “Strong” or “Weak”?

Analyzed the **Language Model Head** (the final layer that predicts the next token)

- All tokens are represented
- $1/m$ Overlap
- Scaling Exponent



The “Chinchilla” Connection

- model size N vs loss

Chinchilla (**empirically**): 0.35

This paper (**theoretically**): 0.4



Too close

**superposition is (likely) the physical
mechanism driving these famous laws.**

Conclusion

intelligence of these models might simply be the result of a high-dimensional packing problem

(They are trying to squeeze the infinite complexity of the world into a finite box, and the math says the bigger the box, the less things get crushed.)

Robustness: We don't need "perfect" data to get scaling laws.

The Limit: The paper suggests that this $1/m$ scaling is a "geometric floor."

Efficiency: encouraging superposition could enable smaller models to match the performance of larger ones (with less superposition) and make training more efficient

Future Architectures: need architectures explicitly designed to handle superposition better